Process Characterization Master Plan

A-Mab Drug Substance — PCMP-001

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ADCC, antibody dependent cell mediated cytotoxicity; AEX, anion exchange chromatography; AMV, analytical method validation; BLA, Biologics License Application; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; LRV, log reduction value; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; PD, Process Development; PPQ, process performance qualification; PRESS, prediction error sum of squares; QbD, quality by design; RSM, response surface methodology; SEC, size exclusion chromatography; SOP, standard operating procedure; UF/DF, ultrafiltration and diafiltration; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

This master plan defines how the A-Mab drug substance process will be characterized before it is validated at the receiving site. A-Mab is a humanized IgG1 monoclonal antibody produced by fed-batch cell culture at a production bioreactor working volume of 15,000 L and purified on the platform train, then concentrated and exchanged into formulation buffer at a drug substance target of 75 g/L. Characterization covers Steps 3–10 of that train, listed with their role in Table 1.1. Every study is executed on a qualified scale-down model at the sending site, and its results are used to justify the ranges the receiving site will operate to.

The campaign is carried by 8 characterization plans and 8 characterization reports, one pair per step. Much of what those plans would state does not change from one step to the next. The quality attribute framework, the risk basis, the approach to scale-down models, the statistical methods and the acceptance framework are common to all of them, and this plan states them once. Each per-step plan then states what is specific to its step and cites this document for the rest. Where a step needs a different rule, its plan will state the rule and the reason for it.

The package this plan sits in is listed in Table 1.2. PTP-001 is the parent. It sets the transfer scope, names the sending and receiving sites and frames the campaign as technology transfer under ICH Q10 (International Council for Harmonisation 2008). RA-001 was performed before any study was executed, and its output is the scope of this campaign (§4). The per-step plans PCP-003 to PCP-010 execute the studies, the matching reports PCR-003 to PCR-010 report them, and PCMR-001 consolidates the campaign for the Biologics License Application.

Four things are outside the scope of this plan. Drug product manufacture and formulation development are covered elsewhere, and this campaign stops at the drug substance. The seed expansion steps that precede Step 3 are not characterized here, because platform experience places no drug substance quality attribute under their control (CMC Biotech Working Group 2009). Validation of the analytical methods is performed under the method validation documents in Table 3.2, and this campaign uses those methods rather than qualifying them. Process performance qualification at commercial scale is Stage 2 work, planned under PTP-001 and executed at the receiving site.

No study result appears in this plan. Each study reports its own results in the report of its step, and PCMR-001 consolidates them.

Table 1.1: The A-Mab drug substance process train and the role of each step in the control strategy.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

Table 1.2: The A-Mab characterization document package.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)

Document ID	Document class	Subject
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

2 Stage 1 characterization strategy

The campaign follows the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012) and the Stage 1 process design activity of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). Under that approach a process is not qualified by showing that fixed set-points reproduce. It is characterized until the mechanism of each step is known well enough to predict the attribute, so that a study can say how much host cell protein the capture pool carries as the protein load on the resin rises and as the elution buffer pH falls, and over what range that relationship holds. The control strategy is then built from those relationships, and the ranges it fixes are ranges the studies have covered.

Prior knowledge is the starting point. A-Mab is produced on a platform that has supported three prior humanized IgG1 antibodies, X-Mab, Y-Mab and Z-Mab, and on which the A-Mab clinical and engineering batches were made (CMC Biotech Working Group 2009). The platform fixes the mechanism of each purification step, and with it the direction each parameter is expected to act in. Protein A binds the antibody through its Fc region while host cell protein and DNA pass in the flow through. The low pH hold inactivates enveloped virus faster as the hold pH falls and as the hold is extended. On the anion exchange resin the counter-ions of the load compete with impurities and virus particles for the quaternary amine ligand, so both host cell protein clearance and virus clearance fall as the load conductivity rises. Characterization measures the size of each of these effects and the range over which it holds, and does not have to establish that the effect exists. Prior knowledge also sets the ranges to be studied, because the platform campaigns have already found the conditions at which the process fails. The low pH hold is not taken below the pH at which the antibody precipitates, and the culture is not extended beyond the point at which falling viability releases host cell protein and DNA into the harvest. Between those bounds the ranges are set wider than the ranges the commercial process will be operated in, so that the studies say something about performance outside routine operation and support later movement within the design space.

The risk assessment comes second. RA-001 ranks every parameter carried into the assessment on its potential impact on a quality attribute and on its potential to interact with other parameters, and assigns it a study type. Parameters that can affect an attribute and can interact are studied together in a multivariate design. Parameters that can affect an attribute but act independently of the others are studied one at a time. §4 gives the outcome across the train.

The studies come third. Each step with a multivariate design runs a two-level screening design in its screening factors, followed by a response surface design in the factors screening carries forward. The screening design identifies which factors act and where the interactions are. The response surface model is the predictive model, and the operating region, the proven acceptable ranges and the capability estimate of the step are all derived from it. Steps without a multivariate design are characterized univariately, with verification runs at the edges of the ranges claimed.

The output is a control strategy. Each report classifies the parameters of its step, states the ranges the step will be operated within, and names the in-process controls and the release tests that hold the attributes the step forms or clears. PCMR-001 assembles those contributions into one control strategy for the drug substance, and states the cumulative viral clearance claim across the steps that carry one (International Council for Harmonisation 2023a).

Criticality is treated as a continuum throughout, for attributes and for parameters (International Council for Harmonisation 2023b). An attribute is placed on a scale of criticality with the score that put it there, and a parameter is classified from the size of its demonstrated effect together with the risk that it leaves its range in routine operation.

Stage 1 is executed at small scale. Everything the campaign says about commercial performance rests on the qualification of the scale-down models (§5) and on the ranges actually studied. Confirmation at commercial scale is Stage 2 work. 5 process performance qualification batches are planned at the receiving site under PTP-001, and they are run against the ranges this campaign establishes (U.S. Food and Drug Administration 2011).

3 Critical quality attribute framework

Table 3.1 is the drug substance quality attribute register the campaign works to. It carries 10 attributes with the acceptance criterion for each, the criticality assigned to it, the score behind that assignment and the step at which it is formed or set. Every characterization plan draws its study responses from this register. No study defines an attribute of its own, and the criteria in this table are the drug substance criteria the campaign works to. A step may also carry an in-process limit that is tighter than the drug substance criterion, and §7 states how the two relate.

Criticality was assigned with the risk tool of the case study, in which the score is the impact of the attribute on safety or efficacy multiplied by the uncertainty in that impact (CMC Biotech Working Group 2009). Attributes scored high or very high are called critical, and 6 of the 10 attributes are critical on that convention. 2 of them are ranked between two levels rather than at one, and those are Host Cell Protein (HCP) and Leached Protein A. Each is studied at the higher of the two levels it spans, so a study of either is designed to the acceptance criterion of a critical attribute (International Council for Harmonisation 2023b).

The register separates what the process forms from what it must remove. The glycan attributes, the acidic charge variants and the aggregate level are formed in the production bioreactor. The glycosylation attributes are not modified by the purification train, because neither affinity capture nor the charge-based separations discriminate between glycoforms at platform conditions (CMC Biotech Working Group 2009). Characterization of those attributes is therefore an upstream activity and PCP-003 carries it. Aggregate is formed upstream and reduced at cation exchange, and the acidic variants can increase during the low pH hold, so the plans for those two steps study the change their step makes rather than the level it forms. The process impurities are carried into the purification train by the harvested culture fluid and are cleared by the chromatography steps, so each purification plan studies the clearance its step provides and the input level it is challenged with.

The 2 viral safety attributes are cumulative. Their acceptance criteria in Table 3.1 are requirements on the whole process, and no single step is required to meet them alone. The low pH hold disrupts the lipid envelope of an enveloped virus, so it inactivates XMuLV and carries no claim against the non-enveloped parvovirus MVM. The anion exchange resin binds virus particles by their net negative surface charge while the antibody flows past. The virus filter retains particles on a membrane whose pore size rating sits below the diameter of MVM. Each of those mechanisms acts on a different property of the particle, its envelope, its charge and its size, so surviving one does not make a particle more likely to survive the next and the log reduction values may be added (International Council for Harmonisation 2023a). Each of those three plans states the step contribution it must demonstrate, back-calculated from the cumulative requirement less the nominal contribution of the other steps (§7). PCMR-001 states the cumulative claim, and no per-step report claims the whole of it.

Every attribute in Table 3.1 is measured by a validated method, and every study is executed under a controlled procedure. Table 3.2 lists the procedures and method validations cited across the campaign. Each plan cites the subset that applies to its step and states the precision of the methods its responses depend on, because the precision of an assay sets the smallest effect a study of that response can resolve.

Table 3.1: The A-Mab drug substance quality attribute register, with the step at which each attribute is formed or set.

CQA	Category	Acceptance	Criticality	Tool #1	Formed or set at
Afucosylation	glycosylation	2–13 % of glycans	H	60	Production Bioreactor
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48	Production Bioreactor
High Mannose	glycosylation	3–10 % of glycans	VH	80	Production Bioreactor
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60	Production Bioreactor
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4	Production Bioreactor
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36	Production Bioreactor
Residual DNA	process impurity	0–0.001 ng/dose	VL	6	Production Bioreactor
Leached Protein A	process impurity	0–5 ppm	L-M	16	Protein A Chromatography
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20	Low-pH Viral Inactivation
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20	Anion Exchange Chromatography

Table 3.2: Controlled procedures and analytical method validations cited across the campaign.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP

Reference	Title	Type
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2012	Operation and Post-Use Integrity Testing of the Small-Virus Retentive Filtration Step	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

Reference	Title	Type
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation

4 Risk-based prioritization

RA-001 was completed before any characterization study was executed, and its output is the scope of this campaign. It is not a classification of parameters. A parameter is classified only after the study that measures its effect has reported, under SOP-4001 and in the report of its own step (§7).

The method is risk ranking and filtering (CMC Biotech Working Group 2009; International Council for Harmonisation 2023b). Each parameter carried into the assessment was ranked twice. One ranking is the potential impact of a deviation in that parameter on a quality attribute, weighted by the criticality of the attribute affected. The other is the potential for the parameter to interact with other parameters of the same step. Where no data or rationale supported an assessment, the parameter was ranked at the highest level. The combined score then decides the study type. A high score with interaction potential sends the parameter into a multivariate design, a high score without it into a univariate study, and a score below the filter into no new study at all.

Table 4.1 gives the outcome across the train. 37 parameters are carried into the campaign, of which 22 are assigned to a multivariate design and 15 to univariate assessment. Parameters that fell below the filter are not listed in this plan. They are operated to the ranges established from platform manufacture, and RA-001 records them together with the reasoning that excluded them.

The split is uneven across the train, and it follows what each step does. The production bioreactor carries the largest multivariate design because it forms most of the quality attributes in Table 3.1, and because its parameters reach those attributes through one culture environment. Carbon dioxide accumulation acidifies the medium, the base added to hold pH at set-point raises osmolality, and pH and osmolality act together on the Golgi enzymes that attach galactose and fucose, so a change in gassing arrives at the glycan through three routes at once. Culture duration acts on the same enzymes for longer, and platform experience is that a longer culture lowers afucosylation (CMC Biotech Working Group 2009). Parameters that reach one attribute by several routes cannot be separated one at a time, which is why they are studied together. 6 of the 8 steps have a multivariate design. The 2 that do not are harvest and clarification, which recovers and clarifies the culture fluid without forming a product quality attribute, and the ultrafiltration and diafiltration step, which concentrates the pool and exchanges it into formulation buffer. The parameters of those two steps are assessed one at a time, and their plans present no designed experiment.

The assignment is prospective, and it may be wrong. If a study shows an effect where the assessment expected none, or a parameter proves to interact where it was assumed independent, the affected plan will be amended before execution continues and the change recorded in the report of that step. RA-001 is revisited once the campaign has reported, so that the risk basis carried into Stage 2 reflects what the studies measured.

Table 4.1: Characterization scope per step, and the plan and report that cover it.

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
3	Production Bioreactor	9	5	4	PCP-003 / PCR-003
4	Harvest / Clarification	3	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	1	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	4	1	PCP-008 / PCR-008
9	Small Virus Retentive Filtration	2	2	0	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration (formulation)	3	0	3	PCP-010 / PCR-010

5 Scale-down model strategy

Every characterization study in this campaign is executed at small scale. What the studies show applies to the commercial process only if the model used is representative of it, so qualification of the scale-down model is the first thing each plan specifies and the first thing each report defends. Qualification is performed under SOP-1001.

The principle is the same at every step. Parameters that do not depend on scale are operated at the commercial set-point in the model. Parameters that do depend on scale are set so that the model matches the commercial equipment on the property through which scale reaches the product, which is volumetric mass transfer and mixing time in a bioreactor, linear velocity and residence time on a column, and flux and transmembrane pressure across a membrane. The model is then run at set-point conditions and its input and output attributes are compared with data from the at-scale process. A model is qualified for an attribute when the difference between scales is smaller than the variability of the method that measures it (§7).

What has to be matched differs by step, and each plan states the basis it uses. For the production bioreactor, pH, temperature, osmolality and culture duration are operated at the commercial set-points, while agitation and gas flow rates are set so that mass transfer and the carbon dioxide accumulation of the small vessel reproduce those of the commercial vessel. For the chromatography steps, bed height, linear velocity, residence time, protein load per litre of resin and the buffer volumes expressed in column volumes are held equal to commercial values, and column efficiency is confirmed by plate count and peak asymmetry before a study is executed. For the low pH hold, the model must reproduce the mixing and the time taken to reach the hold pH, because inactivation depends on the pH the material sees and on how long it sees it. For virus filtration, the volumetric load per membrane area, the flux and the pressure are matched, and the same filter type is used. For ultrafiltration and diafiltration, the membrane chemistry, the channel path length, the flux, the transmembrane pressure and the number of diavolumes are matched.

The three steps that carry a viral clearance claim are qualified twice over. Their models must be representative of the process step, and they must also be suitable for a spiking study under the conditions of ICH Q5A (International Council for Harmonisation 2023a). A virus spike changes the load, so the spiked model is compared with the unspiked model for the attributes the step controls, and the spike volume is kept small enough that the conductivity and pH of the load stay inside the range studied. Where the spiked and unspiked models differ, the difference is reported and the clearance claim is bounded by it.

Every designed study includes replicated centre points at the set-point condition, and those replicates do two jobs. They provide the pure error term that the lack of fit test needs (§6), and they monitor the model across the execution of the design. A drift in the centre point response between the start and the end of a design is treated as evidence about the model or the material and is investigated before the design is analysed.

A model that cannot be qualified for an attribute is not used to make a claim about that attribute. Where this occurs, the plan of the step will state what is claimed instead, which may be control of the attribute by release testing, by an in-process limit, or by data from the at-scale process. The instrumented systems used for the studies are maintained under calibration and change control, and each plan records the systems its study uses.

6 Common statistical approach

The studies are designed and analysed in coded factors. The set-point of a parameter is coded zero and the edges of its characterization range are coded −1 / +1. Coding puts every parameter on the same footing, so an effect is expressed as the change in the response over half the characterization range and effects on different parameters can be compared directly. Every design is randomized in execution order, and the analysis is performed under SOP-1002.

Each step with a multivariate design begins with a two-level design in all of its screening factors, augmented with replicated centre points. A full factorial is used where the number of factors allows it, and a resolution V half fraction where it does not, so that main effects and two-factor interactions remain estimable. The model fitted to a screening design carries the main effects and the two-factor interactions. An effect is twice the coded coefficient and is reported with its p-value at a significance level of $\alpha = 0.05$.

The factors that screening carries forward go into a face-centred central composite design with replicated centre points, and a full quadratic model is fitted to it. A two-level screening design holds each factor at two settings, so it cannot resolve curvature and the magnitudes it returns are refined by the quadratic model before they are used. The design space, the proven acceptable ranges and the capability estimate are predicted from the response surface model, and each report names the model each number came from.

Table 6.1 gives the planned design structure for the 6 steps with a multivariate design. The run counts include the centre point replicates. The response count is the number of measured responses fitted per design, and each response is modelled separately.

Model adequacy is judged on four things. The first is the significance of the model terms. The second is the coefficient of determination together with its adjusted form, which penalises terms that buy no explanation. The third is the predicted coefficient of determination computed from the prediction error sum of squares, which says how the model behaves on data it was not fitted to. The fourth is the lack of fit test, in which the residual variation left by the model is compared with the pure error of the centre point replicates. A model whose predicted coefficient falls well below its adjusted coefficient is over-fitted and is reduced before it is used. A model with significant lack of fit does not describe the data, and the response is then reported without a predictive model and its range set by the observations. Residuals are examined against the predicted values and against run order, and normality is checked on a quantile plot.

A proven acceptable range is computed for each parameter and each attribute the step forms or clears, in two analyses. The first holds the other parameters at the centre of the characterization range and finds the range of the parameter over which the predicted response meets its criterion. The second propagates the variation of the other parameters within their normal operating ranges by Monte-Carlo simulation, and finds the range over which the predictive bound still meets the criterion. The second analysis is the one carried into the control strategy, because the routine process varies every parameter at once. Each report presents both, so that the cost of the surrounding variation can be read directly.

Capability at commercial scale is estimated by simulation. 2,000 batches are simulated with every parameter drawn about its set-point inside its normal operating range, and the drug substance attributes of each simulated batch are predicted from the characterized process. Capability is then computed one-sided against the criterion that applies. For an impurity the index is computed against the ceiling, for a viral clearance attribute against the floor, and for an attribute with a target range as the smaller of the two. The estimate is conditioned on the fitted models and on the normal operating ranges used to generate it, so narrowing a normal operating range raises the index and widening one lowers it.

Table 6.1: Planned design structure for the steps with a multivariate design.

Step	Unit operation	Design	Factors	Runs	Centre points	Responses
3	Production Bioreactor	Two-level screening	5	19	3	5
3	Production Bioreactor	Face-centred central composite	4	28	4	5
5	Protein A Chromatography	Two-level screening	4	19	3	3
5	Protein A Chromatography	Face-centred central composite	4	28	4	3
6	Low-pH Viral Inactivation	Two-level screening	3	11	3	3
6	Low-pH Viral Inactivation	Face-centred central composite	3	18	4	3
7	Cation Exchange Chromatography	Two-level screening	4	19	3	3
7	Cation Exchange Chromatography	Face-centred central composite	4	28	4	3
8	Anion Exchange Chromatography	Two-level screening	4	19	3	4
8	Anion Exchange Chromatography	Face-centred central composite	4	28	4	4
9	Small-Virus Retentive Filtration	Two-level screening	2	7	3	3
9	Small-Virus Retentive Filtration	Face-centred central composite	2	12	4	3

7 Acceptance criteria framework

This section states what each per-step plan must decide and on what basis it decides. A numeric threshold is stated here only where it is fixed for the whole campaign. Where a threshold depends on the step, on the attribute or on the precision of the method, the criterion is defined in the plan of that step and the reason is given there.

A scale-down model is qualified when the attributes it produces at the set-point condition agree with data from the at-scale process, and when the difference between scales is smaller than the variability of the method that measures the attribute. Each plan lists the attributes compared, the source of the at-scale data and the number of model runs used. Where at-scale data do not exist for an attribute, qualification rests on engineering equivalence and the plan states that this is the basis.

A response surface model is accepted as predictive when its terms are significant at $\alpha = 0.05$, when the lack of fit is not significant against pure error, and when the predicted coefficient of determination is consistent with the adjusted coefficient. The thresholds applied to the coefficients of determination are stated per study, because what counts as an adequate fit depends on the precision of the assay that measures the response. A study whose response is measured by a low precision method may be adequate at a lower coefficient than one measured by a precise method, and the plan says which case it is in.

An operating region is acceptable when every attribute the step forms or clears meets its criterion across the whole region while the other parameters vary within their normal operating ranges. The criterion a response is judged against is the in-process limit of the step where one has been established, and the drug substance acceptance criterion where none has. An in-process limit is generally the tighter of the two, because it is set with margin against the clearance the downstream steps must still deliver, so an operating region judged against it is smaller than one judged against the drug substance criterion. For a viral clearance response the criterion is the step contribution back-calculated from the cumulative requirement less the nominal contribution of the other clearing steps.

Three ranges are used throughout the campaign. The normal operating range is the range the commercial process is operated within, and it is set by the control capability of the equipment. The characterization range is wider, and it is the knowledge space the studies cover. The proven acceptable range is the range over which an attribute has been shown to meet its criterion. A proven acceptable range is not extrapolated past the edge of the characterization range, and where the acceptable region runs to that edge the report states that the range is bounded by the study and not by the process. The design space is the multivariate region inside the knowledge space over which the attributes the step forms or clears meet their criteria, and the normal operating range sits inside it.

A parameter is classified under SOP-4001 after the study that measured its effect has reported. A parameter whose variation affects a critical quality attribute is a critical process parameter. Where the risk that it leaves its acceptable range in routine operation is low, because it is readily controlled and the acceptable range is wide against the control capability, it is classified as a well controlled critical process parameter (CMC Biotech Working Group 2009). A parameter that affects process performance or consistency without affecting a quality attribute is a key process parameter, and a parameter that affects neither is a general process parameter. No classification appears in a plan, and every classification in a report carries the effect and the control capability that produced it.

No single capability index is set as an acceptance criterion for this campaign. Capability is estimated per attribute and reported with the margin between the simulated distribution and the criterion, and it is read together with the proven acceptable ranges. The acceptance decision for a step rests on the ranges the study demonstrated, and the capability estimate says how much room the process has inside them.

Any departure from an approved plan is a deviation and is recorded. The report of the affected step states what happened, the magnitude of the departure, the investigation, the impact assessment and the disposition. Where an investigation shows that a dataset was not representative of the process, the affected design is re-executed on requalified material and the operating region is derived from the re-executed data. The superseded dataset is retained and may be referenced to confirm the root cause, and it is not used to define a range.

8 Register of characterization plans

Table 8.1 lists the 8 plans this document parents, with the step each covers, the study types its parameters are assigned to and the report that will deliver its result. Each plan is approved before its study is executed, and each is written against the framework in §3 to §7.

A change to an approved plan is made by amendment, with the reason recorded. A change that alters the scope RA-001 set, or any of the common elements stated in this document, amends this plan as well.

Table 8.1: Register of the per-step characterization plans and their reports.

Plan	Subject	Characterization design	Report
PCP-003	Production Bioreactor (Step 3)	Screening, response surface and univariate	PCR-003
PCP-004	Harvest and Clarification (Step 4)	Univariate and qualitative	PCR-004
PCP-005	Protein A Chromatography (Step 5)	Screening, response surface and univariate	PCR-005
PCP-006	Low-pH Viral Inactivation (Step 6)	Screening, response surface and univariate	PCR-006
PCP-007	Cation Exchange Chromatography (Step 7)	Screening, response surface and univariate	PCR-007
PCP-008	Anion Exchange Chromatography (Step 8)	Screening, response surface and univariate	PCR-008
PCP-009	Small-Virus Retentive Filtration (Step 9)	Screening and response surface	PCR-009
PCP-010	Ultrafiltration / Diafiltration (Step 10)	Univariate and qualitative	PCR-010

9 Deliverables and schedule

The campaign delivers 8 characterization plans, 8 characterization reports and one master report. PCMR-001 consolidates the reports, states the consolidated quality attribute outcomes, the parameter classification across the train and the cumulative viral clearance claim, and provides the control strategy that PTP-001 carries into Stage 2.

The sequence is fixed by dependency and not by date. RA-001 must be approved before a plan is written, because the factors of a design come from its ranking, and a factor added after the design has been executed changes the runs and forces the study to be repeated. A plan must be approved before its study is executed. Studies are executed in train order where material dependency requires it, because the load for a downstream step is produced by the step above it, and a characterization study of cation exchange needs a representative Protein A pool. Steps whose feed can be prepared independently of the campaign may be executed in parallel. Each report is issued when its study is complete and its data have been reviewed, and PCMR-001 is written last.

Process performance qualification follows the campaign at the receiving site and is planned under PTP-001. 5 qualification batches are planned. The ranges in the batch records for those batches come from the design spaces and proven acceptable ranges this campaign establishes, so no qualification batch may be executed until the report of every step that constrains its parameters has issued.

This plan carries no dates. The programme schedule is maintained under PTP-001, and the order of execution follows the material dependencies stated above.

10 Approvals

The approval record for this plan is Table 1. Approval by each function named there confirms that the strategy, the risk basis, the scale-down model approach, the statistical approach and the acceptance framework are acceptable for the campaign, and that the per-step plans may be written against them. An amendment to this plan is approved by the same functions.

References

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